

MÉTODOS MATEMÁTICOS EN BIOINGENIERÍA

TEST # 8

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- P1.** Adapt the Python code that you have studied in order to find the solutions $u(x, t)$ for the following initial and boundary value problem for the diffusion of heat in a bar

$$\frac{\partial u}{\partial t}(x, t) = \frac{\partial^2 u}{\partial x^2}(x, t) \quad \text{para todo } 0 < x < 5, \quad t > 0,$$

with the following boundary conditions:

$$\frac{\partial u}{\partial x}(x, t) = 0 \quad \text{for all } t, \text{ when } x = 0 \text{ and when } x = 5.$$

and the following initial temperature distribution:

$$u(x, 0) = \begin{cases} 5(9 - x^2) & \text{if } 0 < x \leq 3 \\ x^3 - 3^3 & \text{if } 3 < x < 5 \end{cases}$$

Specifically, the code must:

- (a) **[12 points]** Generate `.gif` animations for the evolution in time of the temperature profile $u(x, t)$
- (b) **[12 points]** Generate a figure for the spectrum
- (c) **[12 points]** Compute what will be the temperature of the bar at $x = 4.6$ and $t = 0.11$.

Also, use the code and the `.gif` animations in order to estimate (with three decimal digits precision) the time at which the temperature at $x = 4.6$ is of 63° degrees. **[24 points]**